CHAPTER 9 : Gauss Elimination

Gauss elimination : combine eqⁿs to eliminate unknown

Solving Small # of Eqⁿs.

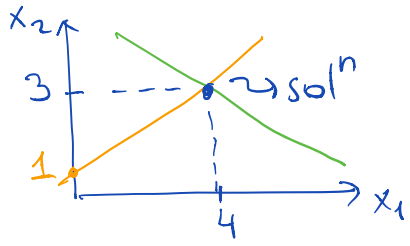
Graphical Method

Ex

$$3x_1 + 2x_2 = 18$$
$$-x_1 + 2x_2 = 2$$

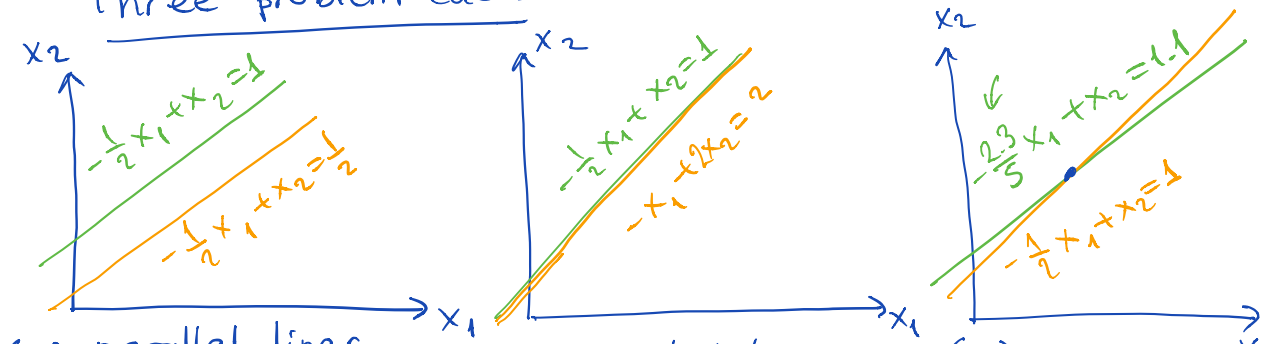
$$x_2 = -\frac{3}{2}x_1 + 9$$
$$x_2 = \frac{1}{2}x_1 + 1 \quad \text{(slope, intercept)}$$

→ lines in 2D



* 3 eqⁿ in 3 unknowns ⇒ planes in 3D.
* Beyond 3 dims. graphical methods are not useful

three problem cases:



(a) parallel liner
no solution

(b) coincident lines
infinite solutions

singular cases

(c) very close to being singular
ill-conditioned
- difficult to identify the exact pt. at which they intersect

Determinants and Cramers' Rule

- Best suited to small # eq^{n}s.

$\underline{\underline{A}}\,x = \underline{b}$

Determinant

Coefficient matrix $\underline{\underline{A}} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Determinant of this system:

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\underbrace{\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}}_{\text{minors}} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\tilde{D} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

EX Compute the determinants of the three systems given in previous example.

(b) $D = \begin{vmatrix} -\frac{1}{2} & 1 \\ -1 & 2 \end{vmatrix} = -\frac{1}{2}(2) - (-1) = 0$

(a) $D = \begin{vmatrix} -\frac{1}{2} & 1 \\ -\frac{1}{2} & 1 \end{vmatrix} = 0$

} Singular systems.

(c) $D = \begin{vmatrix} -\frac{2.3}{5} & 1 \\ -\frac{1}{2} & 1 \end{vmatrix} = -\frac{2.3}{5} \cdot 1 + \frac{1}{2} = -0.04$

almost zero.

Cramer's Rule

$\underline{\underline{A}}\,\underline{x} = \underline{b}$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{D}$$

Ex Use Cramer's rule to solve

$$\begin{aligned} 0.3x_1 + 0.52x_2 + x_3 &= -0.01 \\ 0.5x_1 + x_2 + 1.9x_3 &= 0.67 \\ 0.1x_1 + 0.3x_2 + 0.5x_3 &= -0.44 \end{aligned}$$

$\underline{b}$

$$A = \begin{bmatrix} 0.3 & 0.52 & 1 \\ 0.5 & 1 & 1.9 \\ 0.1 & 0.3 & 0.5 \end{bmatrix}$$

Soln

$$D = 0.3\begin{vmatrix} 1 & 1.9 \\ 0.3 & 0.5 \end{vmatrix} - 0.52\begin{vmatrix} 0.5 & 1.9 \\ 0.1 & 0.5 \end{vmatrix} + 1\begin{vmatrix} 0.5 & 1 \\ 0.1 & 0.3 \end{vmatrix} = -0.0022$$

$$x_1 = \frac{\begin{vmatrix} -0.01 & 0.52 & 1 \\ 0.67 & 1 & 1.9 \\ -0.44 & 0.3 & 0.5 \end{vmatrix}}{D} = -14.9$$

$$x_2 = \frac{\begin{vmatrix} 0.3 & -0.01 & 1 \\ 0.5 & 0.67 & 1.9 \\ 0.1 & 0.44 & 0.5 \end{vmatrix}}{D} = -295$$

$$x_3 = \frac{\begin{vmatrix} 0.3 & 0.52 & -0.01 \\ 0.5 & 1 & 0.67 \\ 0.1 & 0.3 & -0.44 \end{vmatrix}}{D} = 19.8$$

MATLAB : >> det(A) % determinant of matrix A

- For more than 3 eqn's, Cramer's rule is impractical since determinants are time-consuming (by hand or by computer)
- More efficient alternatives are used

Elimination of Unknowns

(1) $a_{11}x_1 + a_{12}x_2 = b_1$

(2) $a_{21}x_1 + a_{22}x_2 = b_2$

$\times a_{21}$

$\times a_{11}$

$a_{21}a_{11}x_1 + a_{21}a_{12}x_2 = a_{21}b_1$

$-\quad a_{11}a_{21}x_1 + a_{11}a_{22}x_2 = a_{11}b_2$

$(a_{21}a_{11} - a_{11}a_{21})x_1 + (a_{21}a_{12} - a_{11}a_{22})x_2 = a_{21}b_1 - a_{11}b_2$

($a_{21}a_{11} - a_{11}a_{21} = 0$)

$$x_2 = \frac{a_{21}b_1 - a_{11}b_2}{a_{21}a_{12} - a_{11}a_{22}} \quad (3)$$

substitute (3) into (1) $\Rightarrow$

$$x_1 = \frac{a_{22}b_1 - a_{12}b_2}{a_{11}a_{22} - a_{21}a_{12}} \quad (4)$$

- Elim. of unknowns is tedious by hand to solve for larger systems.

Naive Gauss Elimination

Goal: extend elimination of unknowns method for larger sets of eq^{n}s by developing a systematic algorithm.

- Systematic technique for forward elimination and back substitution.
- Naive: since it does not avoid division by zero. (→ key, central etc.)

General set of n eq^{n}s:

pivot eq^n

pivot element ← a_{11}

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots & \\ \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Multiply 1st eq^n by $\frac{a_{21}}{a_{11}}$ and subtract from 2nd eq^n

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right] \Downarrow \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a''_{33} & b''_3 \end{array}\right]$$

forward elimination

$\Downarrow$

back substitution

$$x_3 = \underbrace{\frac{b''_3}{a''_{33}}}_{3^{rd}\text{ row}}, \quad x_2 = \underbrace{\frac{b'_2 - a'_{23}x_3}{a'_{22}}}_{2^{nd}\text{ row}}$$

$$x_1 = \frac{b_1 - a_{12}x_2 - a_{13}x_3}{a_{11}} \rightsquigarrow 1^{st} \text{ row}$$

operations

flops: floating-pt operations

Total in Gaussian elimination:

$$\underbrace{\frac{2n^3}{3} + O(n^2)}_{\text{forward elim.}} + \underbrace{n^2 + O(n)}_{\text{back substit.}} \xrightarrow{\text{as } n \text{ increase}} \frac{2n^3}{3} + O(n^2)$$

- Most of the effort is in the elimination step.

Pivoting:

Given:

$$\begin{aligned} 2x_2 + 3x_3 &= 8 \quad \rightarrow a_{11} = 0 \\ 4x_1 + 6x_2 + 7x_3 &= -3 \\ 2x_1 - 3x_2 + 6x_3 &= 5 \end{aligned}$$

- Round-off errors occur if the pivot elemet is close to zero.

Partial pivoting
- Detemine the largest elemet in 1st column.
- Switch the rows so that largest elem is the pivot element

EX: Partial pivoting

Use Gaussian elimination to solve:

$$\begin{cases} 0.0003\,x_1 + 3.0000\,x_2 = 2.0001 \\ 1.0000\,x_1 + 1.0000\,x_2 = 1.0000 \end{cases}$$

Solve with pivot: 0.0003 (very close to zero)

Repeat with partial pivoting

Exact solution: $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$

Soln: Multiply 1st eqn with $\frac{1}{0.0003}$ → a_{21} → a_{11}

$$x_1 + 10000 x_2 = 6667$$

subtract from 2nd eqn:

$$-9999 x_2 = -6666 \Rightarrow x_2 = \frac{2}{3}$$ → substitute into 1st eqn:

$$\Rightarrow x_1 = \frac{2.0001 - 3\left(\frac{2}{3}\right)}{0.0003}$$

very close to each other

Due to subtractive cancellation, the result is very sensitive to the # significant figures

Let $\frac{2}{3} = 0.6666667$ (7 significant figures)

$x_1 \tilde{=} 0.3330000$ → $\frac{2.0001 - 3(0.6666667)}{0.0003}$

e.g. $x_2 = 0.667 \Rightarrow x_1 = -3.33$

3 significant figures

with 15 significant figures: $x_1 = 0.\underbrace{333333\ldots}_{11 \text{ digits}}4037$

Solve with partial pivoting:

$$1.0000 x_1 + 1.0000 x_2 = 1.0000$$

$$0.0003 x_1 + 3.0000 x_2 = 2.0001$$

Multiply 1st eqn 0.0003 and subtract

$$2.9997 x_2 = 1.9998 \Rightarrow x_2 = 0.666\ldots = \frac{2}{3}$$

$$\Rightarrow x_1 = \frac{1 - \frac{2}{3}}{1} = \frac{1}{3}$$ (no round-off error)

Determinant Evaluation by Gauss Elimination:

Determinant of a triangular matrix is the product of the elements in the main diagonal

$$D = a_{11}\, a_{22}\, a_{33} \cdots a_{nn} \quad \Leftarrow$$

Proof for a 3×3 system

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ 0 & a_{33} \end{vmatrix} - a_{12} \underbrace{\begin{vmatrix} 0 & a_{23} \\ 0 & a_{33} \end{vmatrix}}_{0} + a_{13} \underbrace{\begin{vmatrix} 0 & a_{22} \\ 0 & 0 \end{vmatrix}}_{0}$$

$$= a_{11}(a_{22}\, a_{33})$$

- Forward elimination step of Gaussian elim. results in an upper triangular system.

with partial pivoting:

- Determinant changes sign every time a row is switched:

$$D = a_{11}\, a'_{22}\, a''_{33} \cdots a_{nn}^{(n-1)} \, (-1)^{p}$$

(p) → # times that rows are switched.